

# FLOODSENSE

## AR Flood Susceptibility Mobile Application

### Feature Specifications & Complete System Flow

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# 1. Document Overview

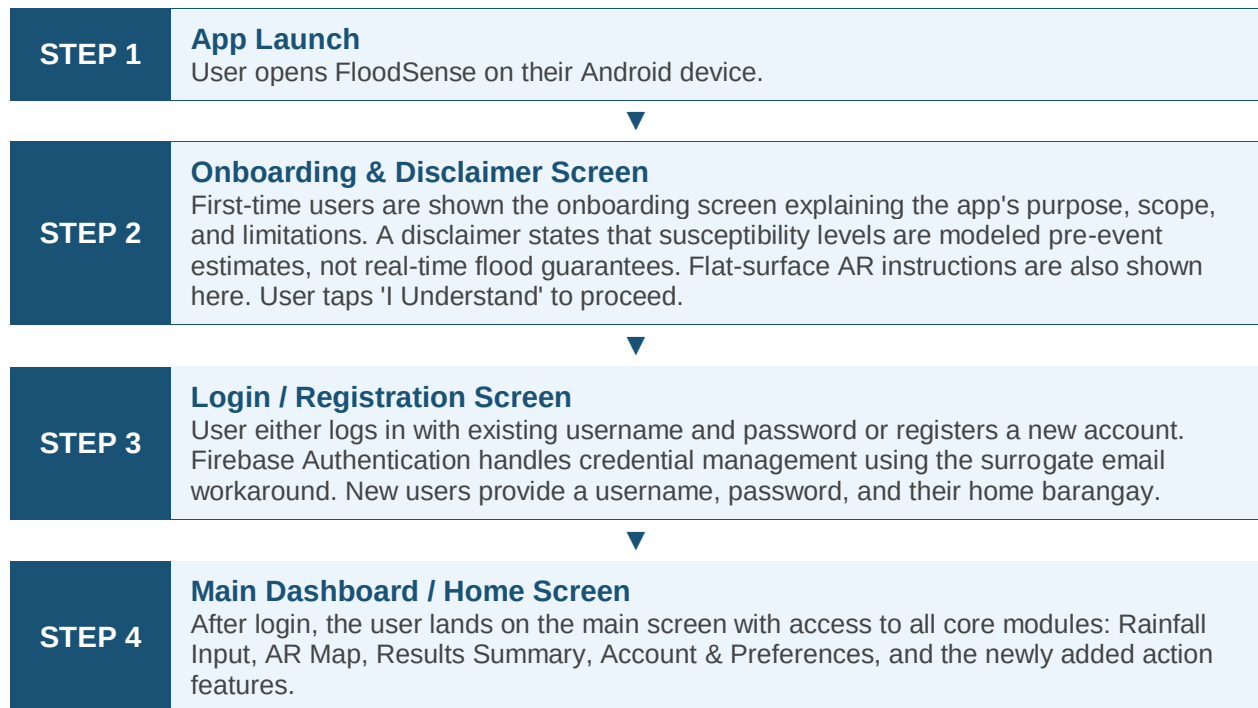
This document details all features of the FloodSense mobile application, including the newly proposed features added in response to the panel's feedback during the thesis defense. It covers the complete user flow from app launch to every available interaction, the purpose of each feature, implementation notes, and feasibility rationale.

FloodSense was originally developed as a pre-event flood awareness tool that combines machine learning-based susceptibility classification with augmented reality visualization. Following the panel's challenge — 'What happens next after users become aware?' — four additional features were designed to transition the system from passive awareness to active, actionable guidance.

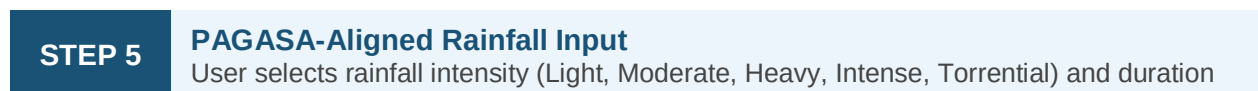
## 2. Complete System Flow

The following is the complete step-by-step flow a user experiences from opening the app for the first time to accessing every feature available in FloodSense.

### 2.1 First-Time Launch Flow



### 2.2 Core Feature Flow (Existing Features)



(1, 3, 6, 12, 24 hours) from plain-language dropdown menus. The app internally maps these to mm/hr values used by the ML model. Saved preferences from a previous session are pre-filled automatically.



#### STEP 6

##### **ML Susceptibility Classification**

The app retrieves the pre-computed susceptibility lookup table from Firebase Firestore based on the selected rainfall intensity and duration combination. No live model inference occurs at runtime. Each barangay is assigned a susceptibility level: Low, Moderate, High, or Very High.



#### STEP 7

##### **AR Visualization**

User taps 'View AR Map'. The camera activates and Vuforia's ground-plane engine scans for a flat surface. The surface detection indicator shows red (invalid) or green (valid). Once a valid surface is confirmed, the color-coded barangay susceptibility map is anchored to the surface as a ground-plane overlay. User can pinch-to-zoom to examine individual barangay labels. A color legend and disclaimer remain accessible during AR view.



#### STEP 8

##### **Results Summary Screen**

A scrollable list of all Bacoor City barangays with their color-coded susceptibility badges is displayed. This provides an at-a-glance overview without requiring the AR interface.

## 2.3 New Feature Flow (Post-Awareness Action Features)

After viewing the AR map or the Results Summary, the following new action buttons are presented to the user, transitioning the app from passive awareness to active guidance.

#### STEP 9

##### **Check My Area (Safety Status Check)**

User taps 'Check My Area'. The app requests GPS permission and detects the user's current location. It cross-references the detected coordinates against a GeoJSON boundary file of Bacoor City's barangays to identify the user's current barangay. The pre-computed susceptibility lookup table is then queried for that barangay under the currently selected rainfall scenario. A plain-language safety status card is displayed.



#### STEP 10

##### **What Should I Do? (Action Checklist)**

Based on the susceptibility level returned for the user's barangay, a tailored action checklist is shown. For Low: monitor and prepare. For Moderate: secure valuables and identify evacuation routes. For High/Very High: proceed to evacuation if advised, avoid flood-prone roads, and contact relatives.



#### STEP 11

##### **Find Nearest Evacuation Center**

Triggered automatically when susceptibility is Moderate, High, or Very High. A ranked list of BDRRMO-designated evacuation centers is displayed, sorted by distance from the user's GPS location using the Haversine formula. Each entry shows the center name, barangay, and estimated distance. User taps 'Get Directions' and is deep-linked to

|  |                                          |
|--|------------------------------------------|
|  | Google Maps for turn-by-turn navigation. |
|--|------------------------------------------|



|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>STEP 12</b> | <b>Share Susceptibility Report</b><br>User taps 'Share Report'. A form appears where the user enters or selects the target barangay (their own or a friend's/relative's) and confirms or adjusts the rainfall intensity and duration. The app writes a report document to Firebase Firestore with a unique ID and generates a shareable URL (e.g., floodsense.web.app/report/abc123). This URL is passed to Android's native share sheet, allowing the user to send it via Viber, WhatsApp, SMS, Messenger, or email. The recipient opens the link in any mobile browser and views the full web-based report. |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 2.4 Supporting Features

|                                  |                                                                                                                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Account &amp; Preferences</b> | <b>User Account Management</b><br>Users can view and update their registered barangay and saved rainfall preferences. Only one set of rainfall preferences is saved per account. Login is username and password only; social login is not supported. |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                   |                                                                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Onboarding (Repeat Access)</b> | <b>Instructions &amp; Disclaimer Revisit</b><br>Users can revisit the onboarding instructions and disclaimer at any time from the main screen via 'View Instructions and Parameters'. |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3. Feature Specifications

#### Feature 1 — AR Flood Susceptibility Map (Existing)

The core visualization feature of FloodSense. Renders a color-coded, barangay-level flood susceptibility map anchored to any detected flat surface through the device camera using Vuforia ground-plane markerless tracking.

| Aspect            | Details                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| Status            | Existing core feature                                                                                                 |
| Technology        | Unity + Vuforia SDK (Ground Plane)                                                                                    |
| Input             | Pre-computed susceptibility lookup table retrieved from Firebase Firestore based on user-selected rainfall parameters |
| Output            | Color-coded AR map overlay: Low (green), Moderate (yellow), High (orange), Very High (red)                            |
| Surface Detection | Green indicator = valid flat surface detected; Red = invalid. Map is locked until surface is confirmed green.         |
| Legend            | Visible within AR view. Full metadata accessible via in-app info panel.                                               |
| Disclaimer        | Accessible from AR view at all times. States susceptibility is a modeled estimate, not a real-time flood guarantee.   |
| Interaction       | Pinch-to-zoom to examine individual barangay labels and susceptibility colors.                                        |
| GPS Used          | No. AR overlay is anchored to a flat surface, not GPS-linked to outdoor position.                                     |

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#### Feature 2 — PAGASA-Aligned Rainfall Input (Existing)

Users specify a rainfall scenario through plain-language dropdown menus aligned with PAGASA's public advisory classification. The app handles all internal mm/hr conversion transparently.

| Aspect            | Details                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| Status            | Existing core feature                                                                                             |
| Intensity Options | Light (<2.5 mm/hr), Moderate (2.5-7.5 mm/hr), Heavy (7.5-15 mm/hr), Intense (15-30 mm/hr), Torrential (>30 mm/hr) |
| Duration Options  | 1 hour, 3 hours, 6 hours, 12 hours, 24 hours                                                                      |
| Design Rationale  | Plain-language labels remove the need for technical meteorological                                                |

|                          |                                                                                |
|--------------------------|--------------------------------------------------------------------------------|
|                          | knowledge. Validated by BDRRMO key informant.                                  |
| <b>Saved Preferences</b> | One set of rainfall preferences saved per user account via Firebase Firestore. |

### Feature 3 — ML Susceptibility Classification (Existing)

A supervised machine learning model trained offline on environmental datasets classifies flood susceptibility per barangay. Results are pre-computed and stored as static JSON lookup tables in Firestore — no live model inference occurs at runtime.

| Aspect                  | Details                                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Status</b>           | Existing core feature                                                                                                                        |
| <b>Algorithm</b>        | Random Forest or XGBoost (scikit-learn / Python)                                                                                             |
| <b>Input Features</b>   | Elevation, historical flood records (BDRRMO), rainfall intensity and duration (PAGASA-aligned)                                               |
| <b>Output Classes</b>   | Low, Moderate, High, Very High — one classification per barangay per rainfall scenario                                                       |
| <b>Runtime Behavior</b> | App retrieves pre-computed JSON from Firestore. No ML server required at runtime.                                                            |
| <b>Data Sources</b>     | BDRRMO historical flood records, CPDCO elevation and land use maps, PAGASA rainfall classification tables                                    |
| <b>Known Limitation</b> | Model trained on static environmental conditions. Does not account for real-time drainage blockages, soil saturation, or river water levels. |

### Feature 4 — Results Summary & Decision Support (Existing)

A scrollable list of all Bacoar City barangays with color-coded susceptibility badges, populated from the pre-computed Firestore lookup table. Provides an at-a-glance overview without requiring the AR interface.

| Aspect             | Details                                                                                |
|--------------------|----------------------------------------------------------------------------------------|
| <b>Status</b>      | Existing core feature                                                                  |
| <b>Technology</b>  | Unity uGUI ScrollView / List                                                           |
| <b>Data Source</b> | Pre-computed Firestore JSON document keyed by selected rainfall intensity and duration |

**Current Limitation**

No sorting, filtering, or search functionality. All barangays displayed in a flat list.

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## Feature 5 — Check My Area: Location-Based Safety Status (NEW)

Addresses the panel's requirement to inform users whether their current location is safe under the selected rainfall scenario. After viewing the susceptibility map, the user taps 'Check My Area' and receives a plain-language safety status card for their current barangay.

| Aspect                | Details                                                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status                | NEW — proposed in response to panel feedback                                                                                                                               |
| Trigger               | User taps 'Check My Area' button after viewing the AR map or Results Summary                                                                                               |
| Technology            | Android Location API (GPS) + GeoJSON barangay boundary file + existing Firestore lookup table                                                                              |
| GPS Usage             | One-time GPS query to detect current barangay. Coordinates are not stored or transmitted — used only for local boundary lookup.                                            |
| Data Privacy          | Compliant with RA 10173. No GPS coordinates are saved to Firebase or any external server.                                                                                  |
| Output                | Plain-language safety status card: barangay name, susceptibility level with color indicator, and brief contextual message.                                                 |
| Example Output (Low)  | 'Your area (Brgy. Zapote) is LOW susceptibility under Heavy rainfall for 3 hours. Stay alert and monitor updates.'                                                         |
| Example Output (High) | 'Your area (Brgy. Talaba) is HIGH susceptibility under Heavy rainfall for 3 hours. Consider taking precautionary action.'                                                  |
| Feasibility           | High. The data backbone (Firestore lookup table) is already in place. GPS-to-barangay detection uses a GeoJSON boundary file — a one-time asset that requires no paid API. |

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## Feature 6 — What Should I Do? Action Checklist (NEW)

Addresses the panel's concern that awareness alone is insufficient and that the system must educate and guide the user's decision-making. A static, level-specific checklist appears after the safety status card, providing concrete pre-event actions the user should take.

| Aspect     | Details                                                                                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status     | NEW — proposed in response to panel feedback                                                                                                                                                             |
| Trigger    | Automatically displayed after the Safety Status card (Feature 5) is shown                                                                                                                                |
| Technology | Static content stored in app or Firestore, mapped to susceptibility levels. No additional logic or API required. A conversational AI assistant capable of answering user queries in the context of their |



|                                  |                                                                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|                                  | susceptibility result is identified as a viable future enhancement, subject to API cost evaluation and institutional approval. |
| <b>Implementation Complexity</b> | Very low. Essentially a content screen with four preset text sets mapped to Low, Moderate, High, Very High.                    |

## Checklist Content by Susceptibility Level

### LOW Susceptibility:

- Monitor weather updates from PAGASA and BDRRMO.
- Prepare your go-bag with essentials (documents, medicine, water, clothes).
- Check drainage around your property.

### MODERATE Susceptibility:

- Secure important documents and valuables in waterproof storage.
- Identify your nearest evacuation center.
- Prepare your go-bag and keep it accessible.
- Inform family members of your location and plan.

### HIGH Susceptibility:

- Proceed to an evacuation center if advised by BDRRMO or barangay officials.
- Avoid flood-prone roads and low-lying areas.
- Bring your go-bag and essential documents.
- Alert relatives and neighbors about your situation.
- Follow all official BDRRMO and PAGASA advisories.

### VERY HIGH Susceptibility:

- Evacuate immediately if not already done.
  - Do NOT attempt to cross flooded roads or waterways.
  - Bring go-bag, documents, medicine, and sufficient food and water.
  - Contact relatives and inform them of your evacuation destination.
  - Follow BDRRMO and PAGASA emergency directives without delay.
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## Feature 7 — Find Nearest Evacuation Center (NEW)

Addresses the panel's requirement to automatically direct users to the nearest evacuation centers. Triggered when the user's barangay is classified as Moderate, High, or Very High susceptibility.

| Aspect                 | Details                                                                                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status                 | NEW — proposed in response to panel feedback                                                                                                                                                    |
| Trigger                | Automatically offered after Safety Status card shows Moderate, High, or Very High result                                                                                                        |
| Technology             | Static evacuation center JSON stored in Firestore + Haversine formula for distance calculation + Google Maps URI deep-link                                                                      |
| Evacuation Center Data | Sourced from BDRRMO as part of existing data-sharing collaboration. One-time data entry of center names, barangays, and GPS coordinates into Firestore.                                         |
| Distance Calculation   | Haversine formula computed on the device using the user's GPS coordinates and each center's stored coordinates. No routing API or paid service required.                                        |
| Display                | Ranked list of nearest evacuation centers showing: center name, barangay, estimated straight-line distance from user.                                                                           |
| Directions             | Each center entry has a 'Get Directions' button that deep-links into Google Maps (via geo: URI or maps.google.com URL). All routing is handled by Google Maps — no custom routing engine built. |
| GPS Usage              | Same one-time GPS fix used in Feature 5. Not stored or transmitted.                                                                                                                             |
| Feasibility            | High. No paid API required. Google Maps deep-link is a standard Android feature. Evacuation center list is a one-time static dataset obtainable from BDRRMO.                                    |

## Feature 8 — Share Susceptibility Report (NEW)

Allows users to generate and share a full susceptibility report for any barangay in Bacoor City — including their own or a friend's or relative's location — via any messaging or social platform. The report is accessible as a mobile-friendly web page that anyone can open in a browser without installing the app.

| Aspect                       | Details                                                                                                                                                                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Status</b>                | NEW — proposed to extend community-level reach                                                                                                                                                                                                                          |
| <b>Trigger</b>               | User taps 'Share Report' button after viewing the AR map or Results Summary                                                                                                                                                                                             |
| <b>Step 1</b>                | A share form appears. User selects the target barangay from a dropdown (can be different from their own location). Rainfall intensity and duration are pre-filled from the current session but can be adjusted.                                                         |
| <b>Step 2</b>                | App writes a report document to Firebase Firestore with a unique auto-generated ID. Document contains: target barangay, susceptibility level, rainfall scenario, action checklist content, nearest evacuation center for that barangay, timestamp, and disclaimer text. |
| <b>Step 3</b>                | App builds a shareable URL: <code>floodsense.web.app/report/{uniqueID}</code>                                                                                                                                                                                           |
| <b>Step 4</b>                | URL is passed to Android's native share sheet. User selects their preferred app (Viber, WhatsApp, SMS, Messenger, email, etc.). No custom messaging infrastructure is built.                                                                                            |
| <b>Step 5</b>                | Recipient opens the link in any mobile browser. The web page reads the Firestore document by ID and renders the full report.                                                                                                                                            |
| <b>Technology (App side)</b> | Firebase Firestore write + Android share intent (standard Android feature)                                                                                                                                                                                              |
| <b>Technology (Web side)</b> | Single HTML + CSS + JavaScript file hosted on Firebase Hosting (free tier). Reads Firestore document by URL ID parameter and renders all report content dynamically.                                                                                                    |
| <b>Document Expiry</b>       | Optional: Firestore documents can be set to auto-delete after 72 hours to manage storage and data hygiene.                                                                                                                                                              |
| <b>Feasibility</b>           | High. Firebase Hosting is included in the existing free-tier Firebase plan. The web report page is a single HTML file requiring no framework. Android share intent is a one-line implementation.                                                                        |

## Web Report Page — Content Shown to Recipient

When the recipient opens the shared link in their browser, they see the following information:

1. FloodSense report header with generation timestamp and disclaimer
2. Target barangay name and city
3. Rainfall scenario used: intensity label and duration
4. Flood susceptibility level with color indicator (Low / Moderate / High / Very High)

5. What You Should Do checklist — tailored to the susceptibility level of that barangay
  6. Nearest evacuation center name, barangay, and an 'Open in Google Maps' button
  7. Full disclaimer: pre-event estimate only, not a real-time guarantee; always follow BDRRMO and PAGASA advisories
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## 4. Consolidated App Flow Summary

The following summarizes the complete end-to-end flow of FloodSense from first launch through all available features.

| #  | Step / Feature                         | What Happens                                                                                                                                                                    |
|----|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>App Launch</b>                      | User opens FloodSense on Android device                                                                                                                                         |
| 2  | <b>Onboarding &amp; Disclaimer</b>     | First-time users read app purpose, scope, flat-surface instructions, and disclaimer. Tap 'I Understand.'                                                                        |
| 3  | <b>Login / Register</b>                | Existing users log in. New users register with username, password, and home barangay.                                                                                           |
| 4  | <b>Main Dashboard</b>                  | Home screen with access to all modules.                                                                                                                                         |
| 5  | <b>Rainfall Input</b>                  | User selects PAGASA-aligned intensity and duration from dropdowns.                                                                                                              |
| 6  | <b>ML Classification</b>               | App retrieves pre-computed susceptibility lookup from Firestore for selected rainfall scenario.                                                                                 |
| 7  | <b>AR Map View</b>                     | Camera activates. Ground-plane detected. Color-coded susceptibility map rendered on flat surface.                                                                               |
| 8  | <b>Results Summary</b>                 | Scrollable barangay list with color-coded susceptibility badges.                                                                                                                |
| 9  | <b>Check My Area (NEW)</b>             | GPS detects user's barangay. Safety status card displayed with plain-language susceptibility result.                                                                            |
| 10 | <b>Action Checklist (NEW)</b>          | Level-specific 'What Should I Do?' checklist shown based on susceptibility result.                                                                                              |
| 11 | <b>Nearest Evacuation Center (NEW)</b> | Ranked list of BDRRMO evacuation centers sorted by distance. 'Get Directions' deep-links to Google Maps.                                                                        |
| 12 | <b>Share Report (NEW)</b>              | User selects target barangay and rainfall parameters. App generates unique report URL. Android share sheet sends via any messaging app. Recipient views full report in browser. |
| 13 | <b>Account &amp; Preferences</b>       | User manages registered barangay and saved rainfall preferences.                                                                                                                |
| 14 | <b>Revisit Onboarding</b>              | User can re-read instructions and disclaimer from main screen at any time.                                                                                                      |

## 5. New Features Feasibility Summary

| Feature                               | Complexity | Est. Dev Time | Key Dependency                                             |
|---------------------------------------|------------|---------------|------------------------------------------------------------|
| <b>Check My Area (Safety Status)</b>  | Low-Medium | 1 week        | GeoJSON barangay boundary file + existing Firestore lookup |
| <b>Action Checklist</b>               | Very Low   | 2-3 days      | Static content only, no new API or data source required    |
| <b>Find Nearest Evacuation Center</b> | Low-Medium | 1 week        | Evacuation center list from BDRRMO + Google Maps deep-link |
| <b>Share Susceptibility Report</b>    | Medium     | 1.5-2 weeks   | Firebase Hosting + single HTML web report page             |

All four new features operate within the existing Firebase free tier and require no additional paid APIs or services. Combined estimated development time for all new features is approximately 4-5 weeks, which is well within an 8-month development timeline.

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### FloodSense Feature Specifications Document

Department of Computer Studies | Cavite State University — Bacoor City Campus | May 2027